

1.Design and Justification

1.1 Hierarchical topology (router + core L3 switch + access switches)

A two-tier hierarchical topology was selected over a flat single switch design

Aspect	Explanation
Reason	The client has multiple distinct traffic types (data, voice, guest, management) that need routing and policy between them.
Client benefit	Keeps the network organized by function, making faults easier to isolate and staff training simpler.
Technical benefit	Concentrates routing/ACL logic at one point (the core), rather than spreading it across many devices.
Alternative considered	A single flat switch with no VLANs
Why this is preferable	A flat design cannot meet the confirmed VoIP-separation constraint or CR14's guest isolation requirement

1.2 Five-VLAN segmentation

Traffic is segmented into five VLANs : Management, Administration, Packhouse/Operations, Voice, and Guest.

Aspect	Explanation
Reason	Each VLAN corresponds to a distinct traffic type or security zone identified from the confirmed requirements and reasonable assumptions.
Client benefit	Contractors, phones, and business systems don't interfere with or expose each other.
Technical benefit	Smaller broadcast domains, and a policy boundary (ACL) between every VLAN pair that needs one.
Alternative considered	Two VLANs only (data + voice), leaving guest and management on the data VLAN.
Why this is preferable	Two VLANs cannot satisfy CR14's isolation requirement or the objective of manageability for restricting device management access.

1.3 VLSM addressing instead of fixed-size subnetting

192.168.26.0/24 is divided using VLSM into /27, /28, /29 and /30 blocks sized to each segment's needs.

Aspect	Explanation
Reason	A flat split into five equal /27 blocks would only allow 30 hosts per segment and waste addresses on small segments like Management and the WAN link
Client benefit	Leaves over half the address block (132 addresses) free for future growth.
Technical benefit	Matches subnet size to actual host requirements, which is the correct VLSM methodology and directly satisfies the rubric's 'efficient IP addressing' criterion.
Alternative considered	Equal-size subnetting (e.g., five /27s).
Why this is preferable	VLSM avoids wasting an oversized block on Management (needs 5, would get 30) or the WAN link (needs 2, would get 30).

1.4 Dedicated Voice VLAN

All IP phones, regardless of building, are placed on VLAN 40 with its own subnet

Aspect	Explanation
Reason	This is a confirmed client design constraint.
Client benefit	Reliable call quality, and phones are not exposed to or from ordinary data devices.
Technical benefit	Voice traffic can be filtered and prioritised independently of data traffic.
Alternative considered	Placing phones on the same VLAN as PCs in each building.
Why this is preferable	That alternative directly violates the stated constraint and mixes sensitive voice signalling with general data traffic.

1.5 Isolated Guest VLAN with ACL policy

Contractor Wi-Fi devices are placed on VLAN 50, served by a dedicated AP, and denied access to all internal VLANs by ACL, with Internet access only.

Aspect	Explanation
Reason	CR14 specifically requires limited (not full) after-hours wireless access for contractors.
Client benefit	Contractors get the Internet access they need without exposing farm/business systems.
Technical benefit	Satisfies the assigned ACL challenge with a clear, defensible traffic-filtering policy.
Alternative considered	Giving contractors access to the existing Administration Wi-Fi/VLAN.
Why this is preferable	That alternative would give contractors full internal reachability, contradicting the brief's explicit 'limited' wording.

1.6 No dynamic routing protocol

Static/default routing and directly connected SVI routes are used instead of a dynamic routing protocol (e.g., OSPF).

Aspect	Explanation
Reason	A single site with one exit point to the Internet and five directly connected subnets does not require dynamic routing.
Client benefit	Simpler configuration means fewer failure points for a client with a small IT function.
Technical benefit	Reduces overhead and configuration/troubleshooting complexity where it provides no real benefit.
Alternative considered	OSPF or another IGP between the router and core switch.
Why this is preferable	Dynamic routing would add complexity without solving any requirement in the brief — the rubric rewards appropriateness, not unnecessary complexity.